

Sample Workflow: AI-Based Student Performance Analysis

Scenario: Transforming basic student academic information, assessment results, attendance, and learning indicators into a clear and organized student performance analysis plan.

Goal: Take a student's academic details and available performance data and turn them into a structured analysis workflow, including a self-review stage to check completeness, unsupported assumptions, and the reliability of the conclusions.

Step 1: Data Structuring & Planning (The Setup)

Goal: Identify the student's performance-analysis requirements and organize the information needed for meaningful evaluation.

Prompt 1: Act as an expert student performance analysis assistant. I am going to give you basic information about a student, including course/year, subjects, assessment scores, attendance information, assignment performance, learning activities, and any available academic details.

Your task: Extract the key performance-analysis requirements and create a structured checklist covering available data, missing information, performance indicators, analysis stages, and areas requiring verification.

[Insert Student Academic Details Here]

Expected AI Output: A structured checklist showing the student's available performance information, missing information, important indicators, and the main stages of the analysis.

Step 2: Performance Analysis Guidance (The Execution)

Goal: Create a clear step-by-step student performance analysis based strictly on the information organized in Step 1.

Prompt 2: Excellent. Now, using only the student information and checklist created in Step 1, prepare a clear step-by-step student performance analysis guide.

Constraints:

- Keep the analysis simple, structured, and in the correct sequence.
- Clearly separate data preparation, performance measurement, subject-wise analysis, trend identification, and recommendations.
- Do not invent scores, attendance percentages, academic rules, benchmarks, or institutional policies.
- Identify missing student information instead of guessing.
- Clearly distinguish observed results from interpretations or recommendations.
- Mention where official academic records or faculty/institution verification is required.
- Include a final checklist that can be used before the analysis is finalized.

Expected AI Output: A practical step-by-step student performance analysis guide based on the student's information, with missing or uncertain requirements clearly identified.

Step 3: Critique and Refinement (The Required Gate)

Goal: Force the AI to evaluate the performance analysis for completeness, accuracy, logical reasoning, and unsupported assumptions.

Prompt 3: Before we finalize this, act as a strict student performance analysis reviewer. Review the analysis you just created against the following checklist:

- 1 Does the analysis match the student's stated academic information and performance indicators?
- 2 Are all available and missing details clearly identified?
- 3 Are the analysis stages presented in a logical order?
- 4 Were any scores, benchmarks, attendance values, academic rules, or institutional policies invented?
- 5 Are uncertain conclusions and data-dependent requirements clearly marked for official verification?
- 6 Are recommendations supported by the information provided rather than assumptions?

First: List your critique and explain where the performance analysis failed the checklist. **Second:** Output the final, corrected student performance analysis workflow that fixes those issues.

Expected AI Output: A clear self-critique followed immediately by a refined and finalized student performance analysis workflow.

Workflow Principle

This workflow demonstrates progression, dependency between steps, and a mandatory evaluation gate. The student performance analysis should not be considered final until it has been reviewed and corrected in Step 3.